AN ESSENTIALLY FINITE TYPE INJECTIVE LOCAL HOMOMORPHISM OUT OF A GEOMETRICALLY UNIBRANCH LOCAL DOMAIN WITH SEPARABLE RESIDUE EXTENSION IS THE LOCALIZATION OF AN ETALE MAP

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Allowed standard prerequisites (editorial).

- The characterisation of unramified ring maps and the structure of an unramified map as a quotient of a standard etale map. invoked at proof lines 31834-31839. Produces the standard etale cover C' and the surjection onto C.
- Geometric unibranchness and the invariance of the number of branches under smooth maps. used at proof lines 31847-31849. Is what makes the standard etale cover a domain, the key step of the argument.
- Nonvanishing of the contraction of a nonzero ideal for a flat localized etale map of domains. invoked at proof line 31851. Produces the contradiction that forces injectivity of B' -> B.

Source reference. Original lemma 02FM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3197–3197. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0395: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4260–4260. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

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Source reference. Original lemma 31702: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31702–31702. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 31596: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31596–31596. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 31566: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31566–31566. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 10533: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10533–10533. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a geometrically unibranch local domain. Let $A \to B$ be an injective local homomorphism of local rings, which is essentially of finite type. If $\mathfrak{m}B$ is the maximal ideal of B and the induced extension of residue fields is separable, then $A \to B$ is the localization of an étale ring map.*

Proof. We may write $B = C_{\mathfrak{q}}$ where $A \to C$ is a finite type ring map and $\mathfrak{q} \subset C$ is a prime ideal lying over $\mathfrak{m}$. By Algebra, Lemma 02FM the ring map $A \to C$ is unramified at $\mathfrak{q}$. By Algebra, Proposition 0395 after replacing C by a principal localization, we can find a standard étale ring map $A \to C'$ and a surjection $C' \to C$. Denote $\mathfrak{q}' \subset C'$ the inverse image of $\mathfrak{q}$ and set $B' = C'_{\mathfrak{q}'}$. Then $B' \to B$ is surjective. It suffices to show that $B' \to B$ is also injective.

Since A is a domain, the rings C' and B' are reduced, see Algebra, Lemma 10533. Since A is geometrically unibranch, the ring B' is a domain, see by Lemmas 31596 and 31566. If $B' \to B$ is not injective, then $A \cap \mathrm{Ker}(B' \to B)$ is nonzero by Lemma 31702 which contradicts the assumption that $A \to B$ is injective. $\square$